

Identifying Variables — Set A

Level 2 · Ages 9-11 · Read each experiment. Name the independent and dependent variables, and two controlled variables.

Name _____ Date _____

Independent variable — the one thing you change on purpose.

Dependent variable — what you measure to see what happened.

Controlled variables — things you keep the same so the test is fair.

- ① Priya lets a toy car roll down a ramp set at heights of 10 cm, 20 cm and 30 cm. She measures how far the car rolls across the floor. She uses the same car and the same ramp every time.

Independent variable: _____

Dependent variable: _____

Two controlled variables: _____

- ② Jack ties a metal washer to a string and lets it swing. He tries strings 20 cm, 40 cm and 60 cm long with the same washer, pulling it back the same distance each time. He counts the swings in 30 seconds.

Independent variable: _____

Dependent variable: _____

Two controlled variables: _____

- ③ An adult pours 200 mL of warm water at 45 °C into each of three identical cups. Omar wraps one cup in foil, one in wool and one in newspaper. After 20 minutes he measures the temperature of the water in each cup.

Independent variable: _____

Dependent variable: _____

Two controlled variables: _____
